

Empirical modeling perspective to ecosystem change

Albertas Dvirnas from Åke Brännström's group

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About me



Born in Lithuania in
1990



Arrived to Sweden
on Erasmus+ in
2014



PhD in Computational Biology
and Theoretical Biophysics at
Lund University 2022



LUNDS
UNIVERSITET

Postdoc in Brännström's
group in Umeå since
October 2025



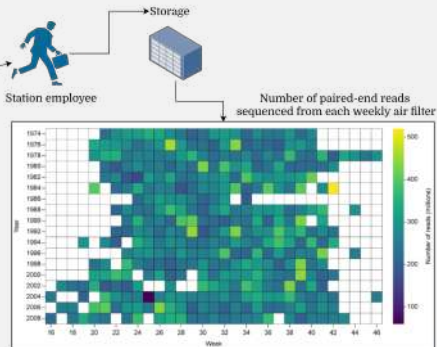
Goals and main questions

- How to model **eDNA** time series and their dynamics using **data-driven** methods?
- Can we successfully forecast into the future?
- How to deal with time series that are multidimensional, noisy, compositional, and possibly missing values?
- Can we find possible causal links between some organisms/environmental cofactors?
- Detect critical changes in ecosystems under stress?

Environmental DNA: station to data



The Kiruna aerosol sampling station



Article

<https://doi.org/10.1038/s41467-025-67676-7>

Airborne eDNA captures three decades of ecosystem biodiversity



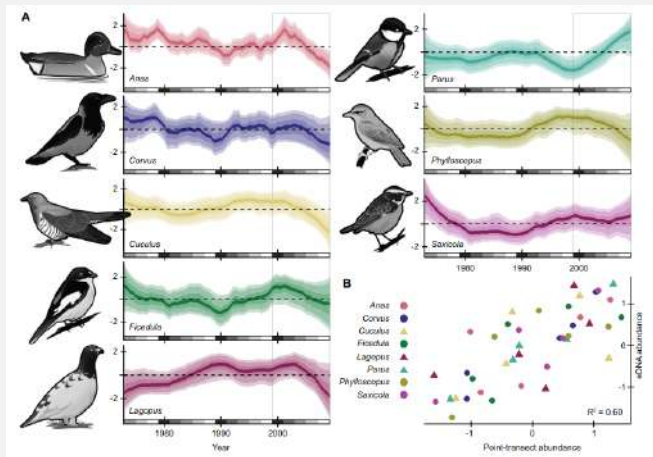
Stenberg's group
in Umeå



icelab

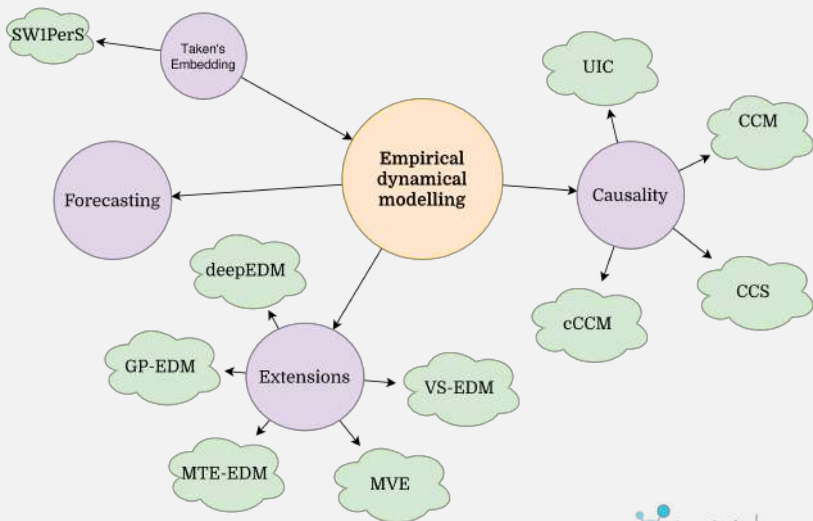
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Concordance of bird eDNA trends and monitoring data

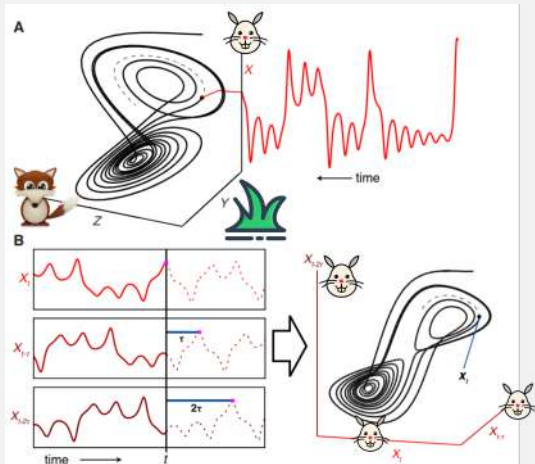


Sullivan, Alexis R., et al. "Airborne eDNA captures three decades of ecosystem biodiversity." Nature Communications (2025).

How to model?

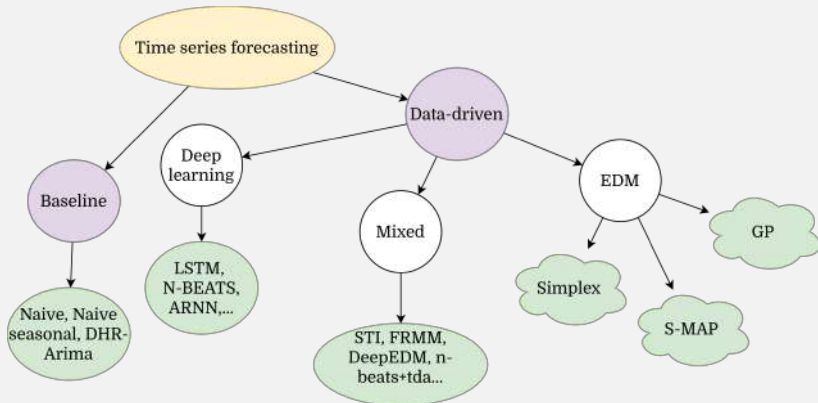


What is EDM?

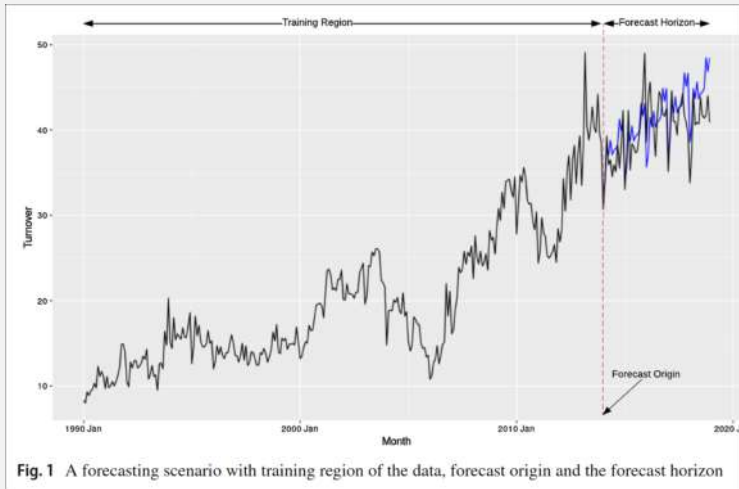


Ye, Hao, et al. "Equation-free mechanistic ecosystem forecasting using empirical dynamic modeling." Proceedings of the National Academy of Sciences 112.13 (2015): E1569-E1576.

How to forecast?

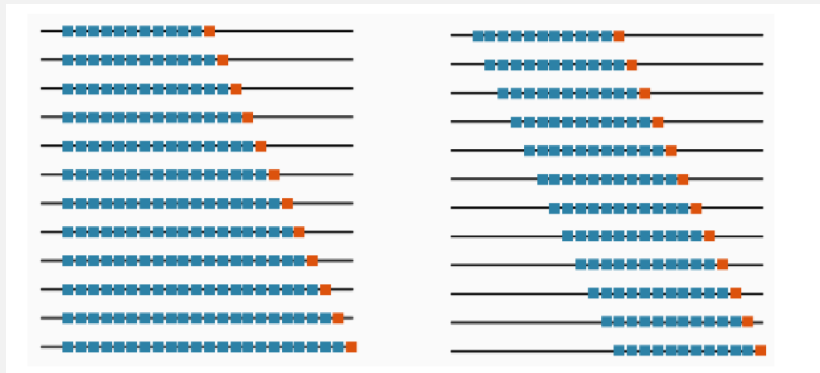


How to forecast?

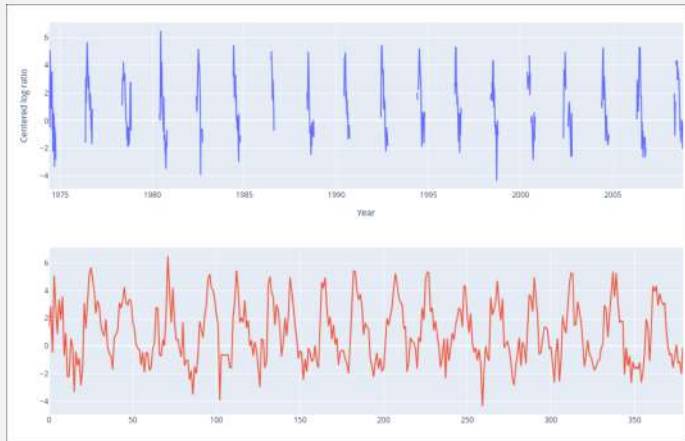


Hewamalage H, Ackermann K, Bergmeir C. Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery*. 2023 Mar;37(2):788-832.

Comparison of forecasting window setups



EDM: eDNA example

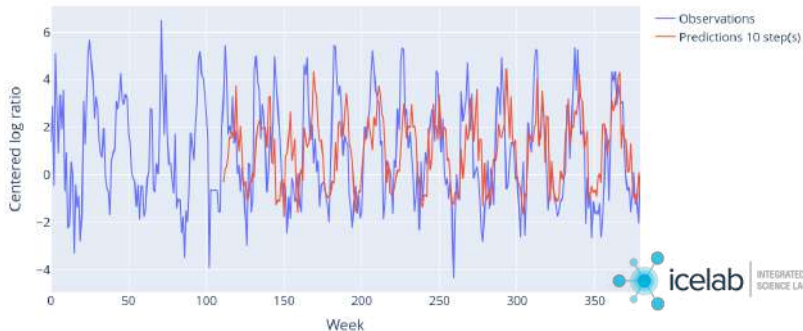


Carex

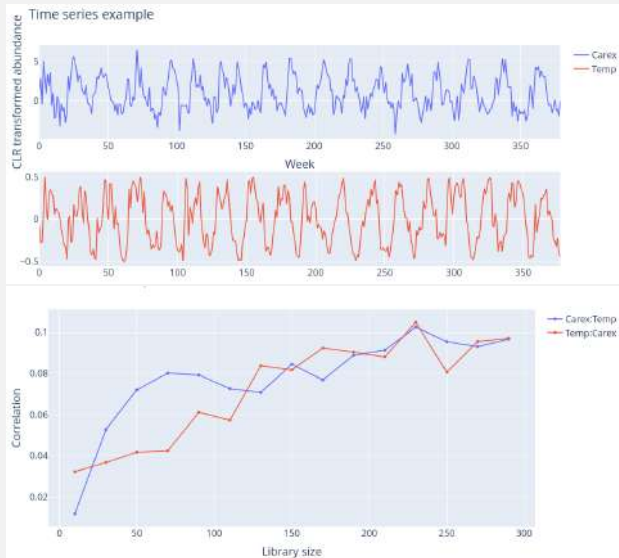
EDM: forecasting example

To run simple EDM simplex forecasting, we need only two parameters: embedding dimension E and lag parameter τ .

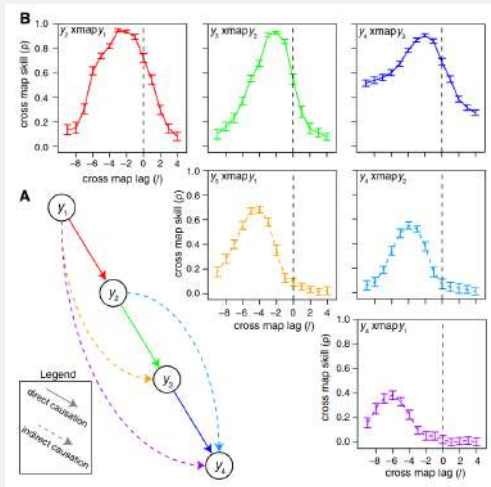
Minimizing mutual information for this time series gives us $\tau = 3$, and a grid search gives us $E = 5$.



Convergent cross mapping example



Causality is complex



Ye H, Deyle ER, Gilarranz LJ, Sugihara G. Distinguishing time-delayed causal interactions using convergent cross mapping. Scientific reports. 2015 Oct 5;5(1):14750.

Outlook/Future goals

- More data: currently only data from Kiruna station, only add odd years and only warm time of the year (total archive contains more than 15000 weekly samples from 6 active stations around Sweden);
- Writing a forecasting methods comparison paper using eDNA as experimental data;
- Using matrix profile approach to deal with some time-series specific challenges;
- Trying to find environmental dependencies for specific organisms using causality detection methods;